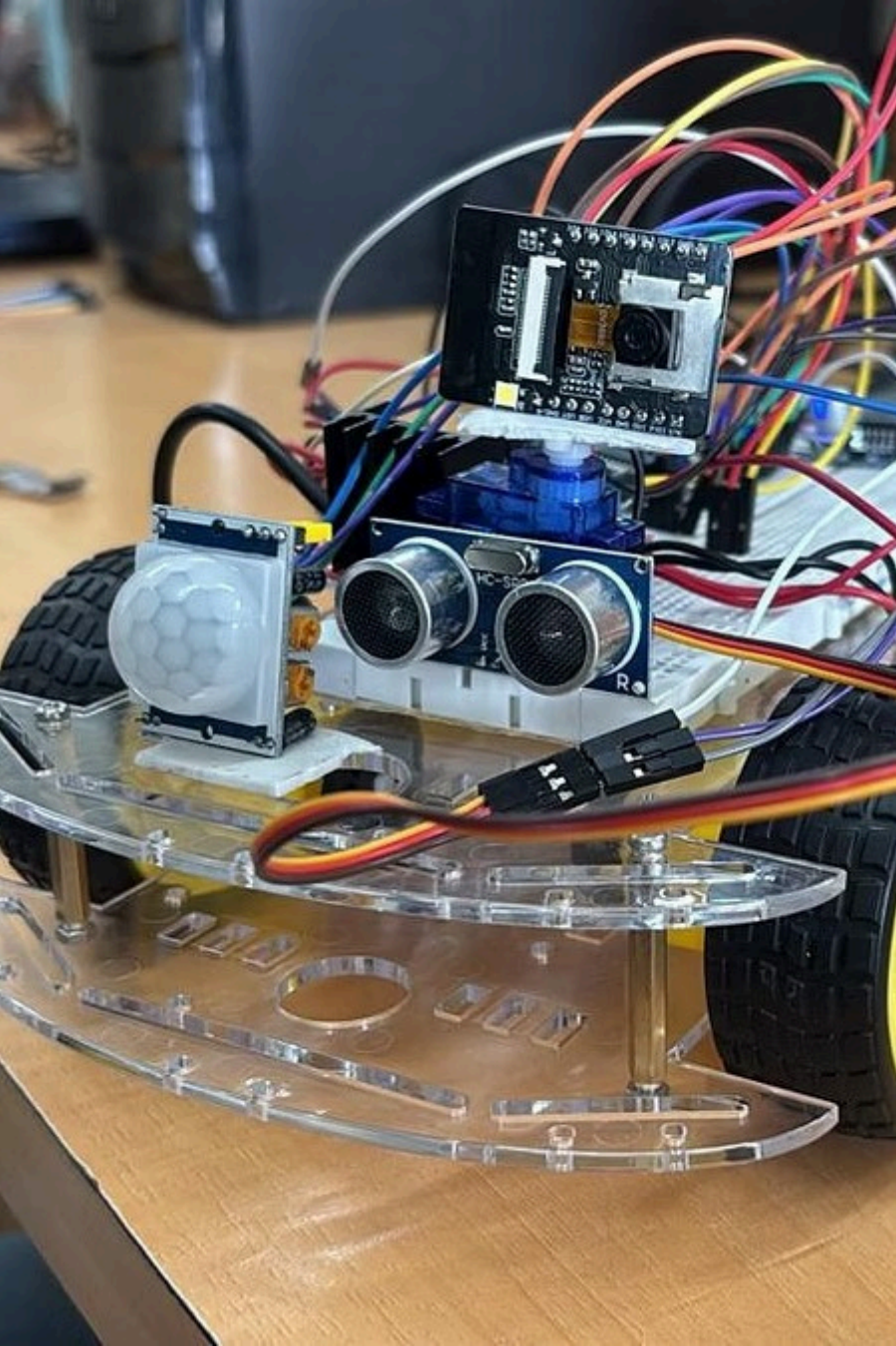




Innovation 2026-Mission : Hardware Autonomous Coastal Surveillance Robot

SMART FORGE CREW

Problem Statement: Design and develop an autonomous mobile surveillance robot capable of monitoring a restricted coastal zone and detecting unauthorized human presence without manual intervention.

TEAM

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Project Objective

1

Accurate Detection

Deploy multi sensor fusion to distinguish humans from environmental noise with minimal false positives.

2

Real-Time Monitoring

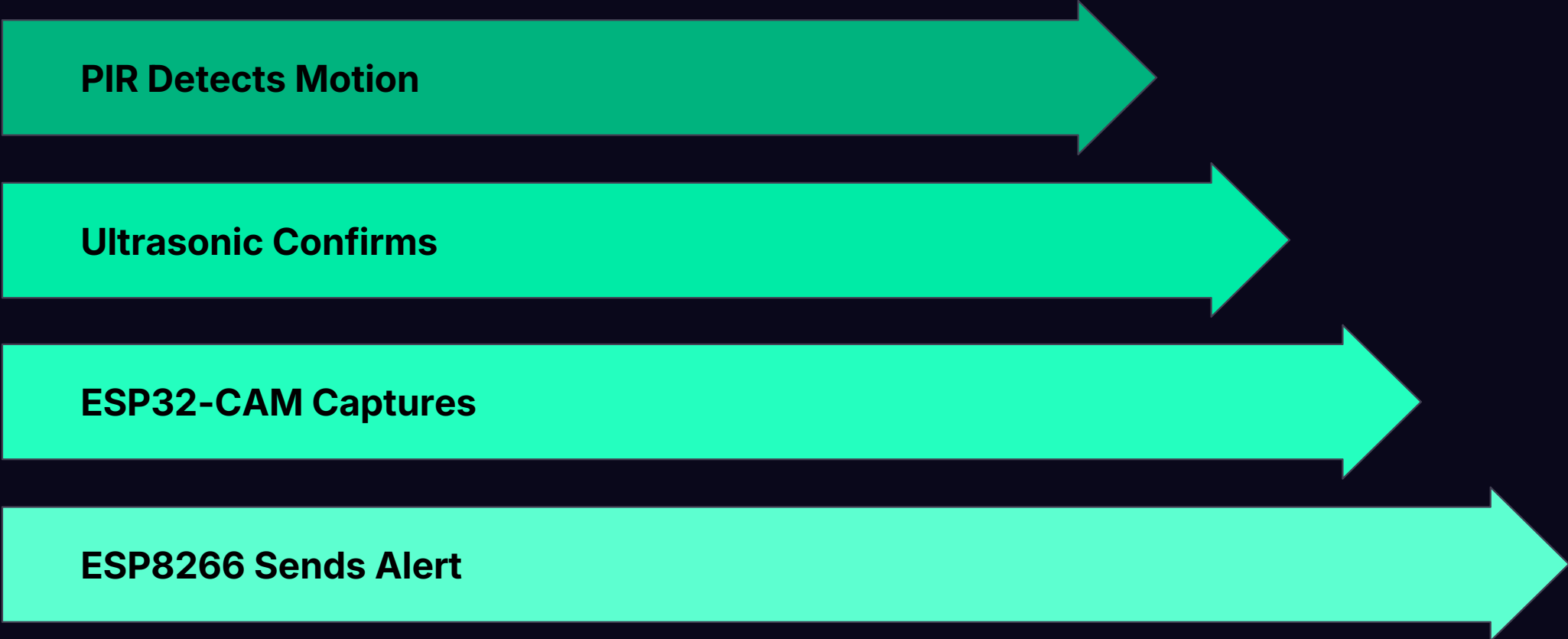
Enable continuous autonomous surveillance with instant image capture and remote alert transmission

3

Optimized Power Usage

Implement adaptive energy management to maximize operational duration on battery power

System Architecture



PIR Detects Motion

Ultrasonic Confirms

ESP32-CAM Captures

ESP8266 Sends Alert

Core Features



Sensor Fusion

Combines PIR and ultrasonic inputs to reduce false detection rates by requiring dual confirmation before triggering alerts.



Smart Image Capture

ESP32 CAM activates only on confirmed detection, balancing surveillance needs with power conservation.



Servo Based Area Scanning

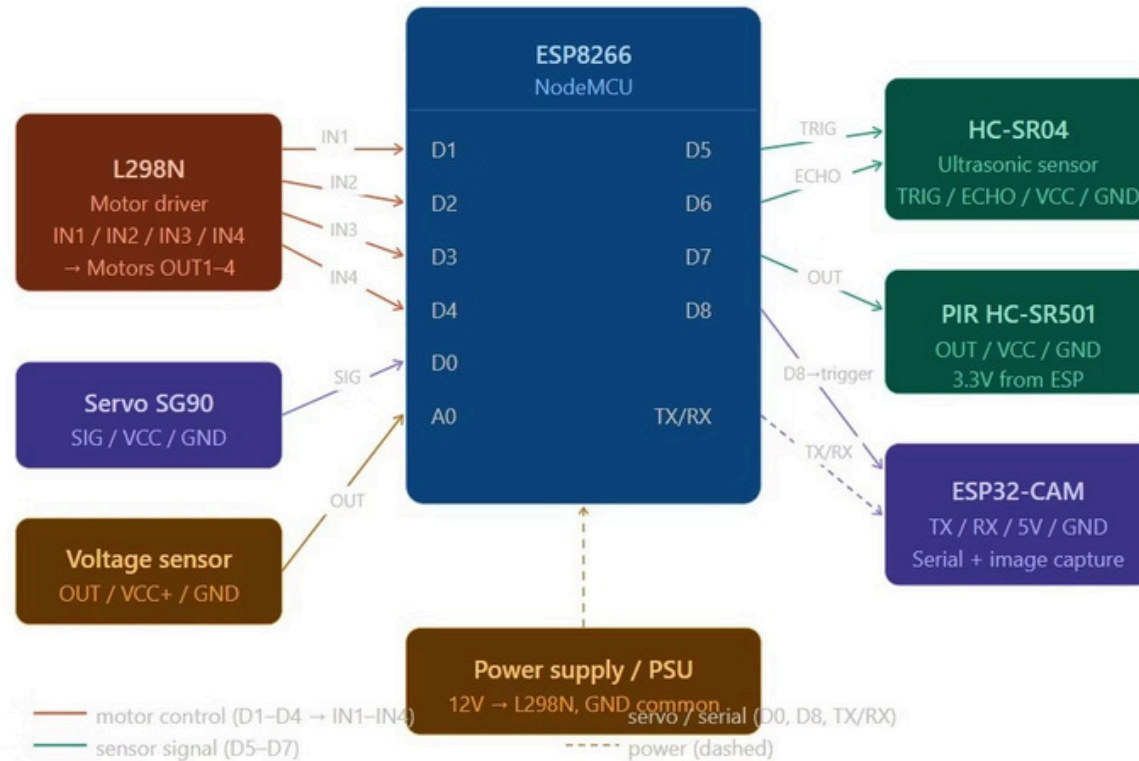
Servo motor scans surroundings and expands field of view which improves detection coverage.



Remote Monitoring

IoT based system enables real time alerts and image transmission to security personnel via web interface

Block Diagram

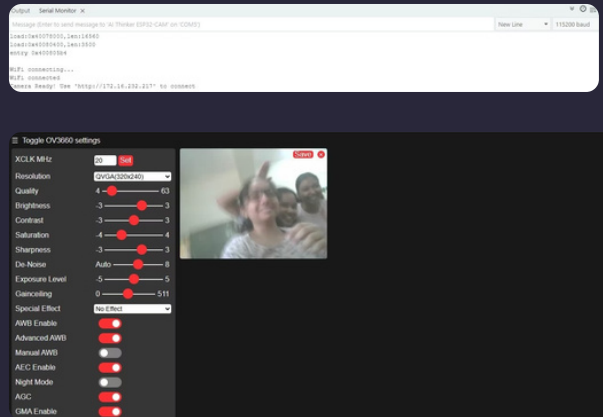


Validation Results

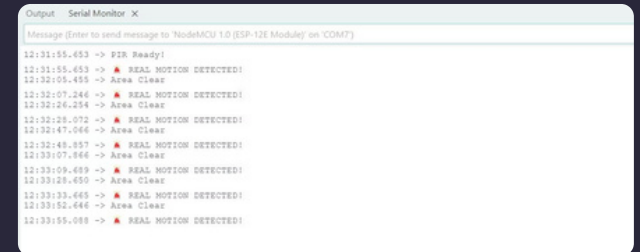
BlynkOutput



Image Capture Verification



Arduino IDE Output



The Challenge

1

WiFi Connectivity Issues

ESP modules depend on stable WiFi.

Signal drops affect monitoring.

2

Integration Complexity

Connecting sensors, motors, camera, and power system together

3

Loose Connections

Multi-strand wires caused unstable connections

Required soldering and proper wiring

Future Enhancements



AI Based Detection

Machine learning models to classify detected objects and reduce false alarms



Battery Regeneration

Using Solar for Battery Regeneration



RFID Usage

Using RFID so that we can differ between Army and Civilian.

Conclusion

The project Robot was successfully developed to achieve the objective of creating surveillance system for defence applications. The system architecture integrates sensor fusion using PIR and ultrasonic sensors, along with the ESP32-CAM for real-time visual monitoring and the ESP8266 NodeMCU for control and communication.

The implementation highlights key features such as accurate intrusion detection, event based camera activation, autonomous movement, and adaptive energy management. During development, challenges including WiFi Connectivity issue, Integration Complexity and Loose Connections were addressed to ensure reliable system performance.

Overall, the system demonstrates a scalable and efficient solution for modern surveillance needs. Future scope includes integration of AI-based threat detection, Solar based battery regeneration and RFID Technology to further improve system intelligence and performance.